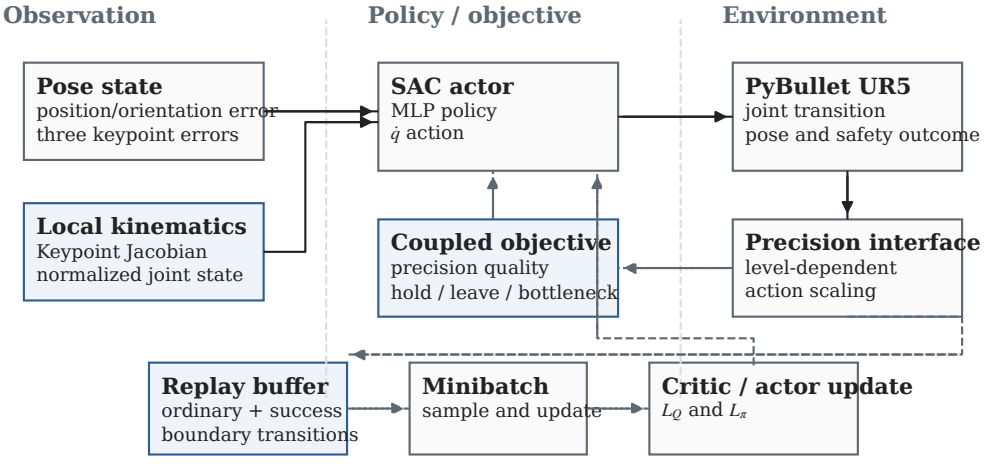
Observation
Policy / objective
Environment
Pose state
position/orientation error
three keypoint errors
Local kinematics
Keypoint Jacobian
normalized joint state
SAC actor
MLP policy
$\dot{q}$ action
Coupled objective
precision quality
hold / leave / bottleneck
PyBullet UR5
joint transition
pose and safety outcome
Precision interface
level-dependent
action scaling
Replay buffer
ordinary + success
boundary transitions
Minibatch
sample and update
Critic / actor update
L_Q and L_π